

Lets write a Book Class

Exercise 1 - Lets Create a class called Book, with the following fields(attributes):

- title (String)
- author (String)
- pages (int)

Steps

1. Create a new project in BlueJ called LibraryLab.
2. Create a class named Book.
3. Enter the following code

```
public class Book
{
    // Fields (attributes)
    private String title;
    private String author;
    private int pages;

    //Constructor and methods to be added
}
```

Can you compile it without error, if so you can move on to adding the Constructor.

Remember a Constructor is a special method that initialises objects as they are created.

Our constructor for Book is going to set up the fields with initial values.

4. Add the following code after the field declaration section

```
// Constructor
public Book()
{
    title = "Unknown";
    author = "Unknown";
    pages = 0;
}
```

Can you compile it without error, try to create a Book object and check the objects state. It should have the values above.

Lets add some methods, *methods implement the behaviour of an object, they provide the functionality*

We are going to add 2 method, one to update the book object and another to print the book details.

```
// Method to print details
public void printDetails()
{
    System.out.println("Title: " + title +
                        ", Author: " + author +
                        ", Pages: " + pages);
}
```

```
// Method to update Book
public void updateBookDetails(String bookTitleNew, String
bookAuthorNew, int numPagesNew)
{
    title = bookTitleNew;
    author = bookAuthorNew;
    pages = numPagesNew;
}
```

Can you compile it? If there are no syntax errors, then create some objects and update the objects with different values, notice that each object as its own value for each of the fields.

5. Lets add getters and setters

- Accessor methods return information about the state of an object. (*getters*)
- Mutator methods change the state of an object. (*setters*)

```
public String getTitle(){
    return title;
}
public void setTitle(String titleNew){
    title = titleNew;
}
```

Can you add the getters and setters for the rest of the fields?

6. **Challenge** Add a (integer) field called bookSales to the Book Class to store the number of sales of the book that have been made. Do you need to make changes to existing methods? Do you need to add new methods?